CODEMITVISHAL · FREE COURSE

Python Pro

From Python Foundations to **job-ready** Python and web development: clean code, testing, concurrency, FastAPI, Django, security, Azure, algorithms and AI.

23
modules (08–30)

100+
runnable lessons

300+
interview answers

0 €
always free

COURSE MAP · 1 OF 2

Core track: Modules 08–20

#	MODULE	YOU'LL BE ABLE TO…
08	**Pythonic code**	write short, idiomatic Python
09	**Data structures**	the right structure, Big-O
10	**Advanced functions**	closures, decorators, recursion
11	**Iterators & generators**	process big data lazily
12	**Advanced OOP**	dunders, MRO, Protocols, SOLID
13	**Errors & logging**	fail safely, log clearly, debug
14	**Packaging**	imports, venvs, a CLI

#	MODULE	YOU'LL BE ABLE TO…
15	**Type hints & code quality**	mypy, ruff, generics
16	**Testing with pytest**	fixtures, mocks, TDD, coverage
17	**Concurrency**	threads, processes, asyncio, the GIL
18	**Data, APIs & databases**	CSV/JSON, HTTP, SQL, REST, time zones
19	**Performance & internals**	memory model, profiling, NumPy
20	**Capstone: budget**	CLI + API + tests

COURSE MAP · 2 OF 2

Web track (21–28), then interviews & AI (29–30)

#	MODULE	YOU'LL BE ABLE TO…
21	**HTTP, Flask & Jinja2**	how the web works
22	**FastAPI in depth**	validation, DI, middleware
23	**SQLAlchemy 2.0 & Alembic**	ORM, N+1, indexes, migrations
24	**Authentication & security**	hashing, JWT, permissions, OWASP
25	**Django + DRF**	ORM, admin, REST APIs
26	**Real-time, tasks, caching**	WebSockets, jobs, Redis
27	**Docker, CI/CD & Azure**	containers, pipelines, cloud deploys
28	**Web capstone**	a secure, tested API

#	MODULE	YOU'LL BE ABLE TO…
29	**Algorithms & data structures**	solve coding-interview problems
30	**AI with Python**	LLM APIs, tools, RAG, agents

Interview prep: Top 100 questions, SQL practice, Git workflow and system design in **interview_prep/**.

HOW EVERY MODULE WORKS

The same 8 steps for every topic

1. **Why**: the real problem this solves
2. **Picture it**: an everyday analogy + a diagram
3. **Try it**: code with its **real output**
4. **Step by step**: what happens on each line
5. **Common mistake**: wrong vs right, side by side
6. **In real jobs**: where you'll meet it at work
7. **Interview**: what to say + the follow-up question
8. **Recap** + a practice task

IN EVERY MODULE FOLDER

`Slides`	slides/modules/Module-NN.pdf
`Lessons`	runnable files with teaching notes
`Exercise`	a starter that fails until you solve it
`Quiz`	QUIZ.md with an answer key
`Interview`	INTERVIEW.md with model answers

HOW TO STUDY

Get the most out of Python Pro

1. Watch or read the module's **slides**.
2. **Run every lesson** and change things to see what happens.
3. Do the **exercise** until it prints **All checks passed!**
4. Take the **quiz**, then answer the **INTERVIEW.md** questions out loud.
5. Open a **Pull Request** with your solution: I review it.

```
# get started
git clone https://github.com/
    CodeMitVishal/python-pro.git
cd python-pro
python -m venv .venv
pip install -r requirements.txt
python 08_pythonic_code/01_comprehensions.py
```